

How LLM Agents Generate Novel Solutions to Hard Problems

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EXECUTIVE SUMMARY

The prevailing account for how agents trained only on pre-existing data produce novel outputs is the phenomenon of emergent abilities: capabilities that are absent in smaller models and surface unpredictably only after model scale crosses a threshold. This framing suggests that novelty emerges less from any special training on innovation than from reaching a critical magnitude of parameters and data, at which previously latent patterns can combine in ways that were inaccessible at smaller scales. Consequently, an agent's apparent creativity reflects the latent structure of its training distribution becoming expressible once the model is large enough.

How that novelty is directed toward hard problems depends on agent architecture. A unified taxonomy decomposes agentic systems into five modules—Perception, Brain, Planning, Action, and Tool Use—each an architectural lever controlling how inference-time compute is deployed and combined across a problem. Rather than relying solely on parametric knowledge, such architectures let an agent search, reason step-by-step, and call external tools to offload work. This reconfiguration of compute over time, not just the stored weights, is what enables an agent to tackle tasks that its base model cannot solve in a single forward pass.

Assessing whether an agent has genuinely solved a hard problem requires passing two distinct gates: the result must be correct, and it must be genuinely new, and the two are not interchangeable. This splits the central novelty question into a before-the-point distinction between real novelty and mere recombination of pre-existing patterns rearranged into a new form. Empirically, agents have shown promise beyond pure mathematics, notably in autonomous chemical research, where a language model paired with domain tools conducted prediction, synthesis planning, and literature search. Yet validating genuine insight against mimicry remains an open challenge.

The flagship Navier–Stokes episode crystallizes these tensions. There an LLM reportedly extended a human-created route to the Millennium Prize problem using heavy computation, yet the claim stayed unconfirmed as of the original reporting. Whether the output was novel insight or an ambitious recombination of existing mathematics remains unresolved, and it raises a broader philosophical question about whether such agents exhibit genuine insight or merely mimicry. The case underscores that for systems trained on existing patterns, proving real novelty—distinct from impressive rearrangement—is precisely as hard as the problems they attempt.

The Navier-Stokes Agent Attempt

The headline claim

The flagship case study in this area is the report that an LLM-based AI system claimed to solve the Navier–Stokes existence and smoothness problem. According to reporting, OpenAI said one of its new AI agents had solved the problem, which is one of the seven Millennium Prize Problems in mathematics [1,2]. The Navier–Stokes problem is described as a 90-year-old math problem that has puzzled experts for decades [1,3]. At its core it concerns establishing the existence of a smooth solution for the motion of an incompressible fluid [1]. The identification of the problem itself reportedly came through an AI assistant: according to ChatGPT, the problem in question is precisely the Navier–Stokes existence and smoothness problem [4]. This was reported against the backdrop of what was described as a year of 'unthinkable' math progress, as increasingly powerful AI systems have pushed into hard mathematical territory [4].

What the agent reportedly produced

The account of the solution describes an AI that took a human-created route to the problem and, using enormous computational resources, pushed it to completion [3]. Framed this way, the achievement was not the model deriving everything from scratch but rather following a human-inspired strategy and completing it at scale [3]. Scientific American noted that, if true, this would mark the second time anyone has solved one of the seven Millennium Prize Problems, and would mean that AI has produced a genuine piece of mathematical work in that domain [5].

A cautious reception

The claim has been met with some caution, however [2]. As of the reporting, independent confirmation was not yet available: one observer noted having been unable to confirm it independently yet [4]. Each new conjecture of this kind has invited scrutiny rather than immediate acceptance [2].

Note

An LLM agent reportedly produced a partial solution to the Navier–Stokes existence and smoothness Millennium Prize problem by extending a human-created route with heavy computation, but the claim remained unconfirmed as of the original reporting [2,3,5].

Mechanisms of Novelty from Patterns

Emergent abilities as the canonical account

The most influential mechanistic account of novelty-from-patterns is the phenomenon commonly called emergent abilities. Emergent Abilities of Large Language Models frames this as an unpredictable phenomenon, in which a given ability is absent in smaller models and appears only once model scale crosses a threshold [6]. Because the capability cannot be predicted by extrapolating the performance of smaller predecessors, the novel-seeming behavior is treated as an emergent property of large-scale pattern-matching rather than a direct training target [6]. Under this account, an agent that seems to discover a new solution is recombining learned patterns past a scale threshold rather than inventing ex nihilo [6].

The interpolation boundary

Yet the same frame fixes a hard limit on how far such novelty can travel. Why Creativity Cannot Be Interpolated argues that AI systems can produce individually novel outputs, but that novelty alone is not creativity [7]. According to that position, genuine creativity must respect the governing constraints of a domain [7]. The operative mechanism is that pattern learning interpolates across a finite training distribution, so recombination can only rearrange what has already been observed [7]. Outputs that violate a problem's structural or mathematical constraints therefore fall outside what interpolation can reach [7].

This recombination account becomes testable on divergent-creativity tasks. A Nature comparison of divergent creativity in humans and large language models reports claims that LLMs are approaching a level of creativity akin to human creativity, evaluated through divergent-creativity metrics [8]. Those tasks reward fluency, flexibility, and originality — exactly the dimensions on which interpolated recombination produces measurable, and also bounded, gains [8].

A fourth mechanism inverts the apparent direction of novelty itself. Cognitive Load Limits in Large Language Models notes that LLMs may hallucinate or produce implausible outputs under conditions of high cognitive load [9]. From this angle, an output that reads as surprising or novel to an observer can be a degradation product of the generative process rather than a genuine discovery [9].

Concrete scientific systems show the recombination mechanism operating productively rather than degeneratively. PyPEF, an integrated framework for data-driven protein engineering, combined machine learning with protein evolution to screen and design proteins with various functions [10]. A generative model proposes candidates from a learned distribution while an evaluation component filters them, so functional novelty emerges from the loop rather than from either model alone [10].

Mechanisms attributed to apparent novelty in LLM outputs

MECHANISM	HOW NOVELTY ARISES	SUPPORTING EVIDENCE
Emergent abilities at scale	A capability absent at small scale appears unpredictably once scale is crossed [6]	Emergent Abilities of Large Language Models [6]
Interpolation and recombination	Rearranges within the finite training distribution; cannot cross structural constraints [7]	Why Creativity Cannot Be Interpolated [7]
Divergent-creativity recombination	Scoreable gains on fluency, flexibility, and originality metrics [8]	Divergent creativity in humans and large language models [8]
Hallucination under load	Implausible outputs mistaken for novel findings under high cognitive load [9]	Cognitive Load Limits in Large Language Models [9]
Generative-evaluation loop	Candidates proposed by a learned model and filtered by an evaluator [10]	PyPEF framework [10]

Note

Novelty is a computational outcome — novel combination of seen patterns; creativity is novelty plus respect for domain constraints, and the two are not interchangeable [7].

Agent Architectures Enabling Novelty

A taxonomy of agentic architectures

The architecture of an agent determines how inference-time compute is deployed and combined across a problem. A unified taxonomy of agentic AI decomposes such systems into five modules — Perception, Brain, Planning, Action, and Tool Use — each an architectural lever [11] that can be pulled to increase the diversity and rigor of generated solutions. The Brain holds the model's knowledge and reasoning, Planning decomposes goals into subgoals, and Action plus Tool Use connect the system to the outside world, letting it write code, run simulations, or search [11] Broadly, agentic LLMs rest on several co-evolving advances — natural language processing, reasoning models, tool integration, reinforcement learning, and agent-based modeling [12]

Chain-of-thought and inference-time compute

The most direct lever for amplifying emergence is spending more reasoning compute at inference time. Emergent abilities — capabilities absent in smaller models that appear sharply once scale is crossed — are now understood to extend beyond traditional LLMs to Large Reasoning Models (LRMs) [13] These LRMs deliberately consume larger inference-time compute budgets and leverage reinforcement learning to reward multi-step reasoning [13] Concretely, the Reward Reasoning Model (RRM) of Guo et al. (2025a) is trained within a reinforcement-learning framework in which the system is rewarded for correct Chain-of-Thought

trajectories [14] This differs from prompt-time chain-of-thought: instead of merely eliciting one explanation, the model is optimized to reason along trajectories that are rewarded, and the reasoning can itself search over possible solution paths [13,14]

Search, self-refinement, and tool-augmented loops

Novel solutions often arrive not from a single generation but from iterative loops. Agentic architectures decompose tasks, invoke external tools, and iteratively refine their outputs across many rounds [15] This does two things at once. First, it turns generation into a search: rather than sampling a single answer, the agent explores many candidate solutions and retains the promising ones [15] Second, it creates a self-refinement loop in which the agent evaluates its own output and reworks failures, the operational heart of reinforcement learning for reasoning [14] Tool use extends the agent beyond its parametric memory: a proposed solution can be checked against a solver, a literature database, or a simulation instead of being trusted on faith, which lets the agent reach conclusions it could not have 'remembered' from training data alone [15]

Amplifying emergence toward hard problems

Taken together, these features explain how inference-time compute and search can amplify emergent behavior in a way that matters for hard problems. The emergence discussed previously relied on scale alone, whereas the agentic architecture adds a structural amplifier. Planning decomposes an intractable problem into checkable subgoals [11] tool use supplies external verification, and inference-time reasoning search explores an exponentially larger space of arguments than a single forward pass [13] Reinforcement-trained reasoning models then bias that search toward trajectories that satisfy the subgoals [14] The result is a system whose search over reasoning paths can, in principle, reach configurations unreachable by pattern recall from training data alone [15]

Note

The architectural picture: scale creates the possibility of emergence; inference-time compute, search, self-refinement, and tool use are the levers that amplify it into non-obvious solutions [13,15].

Assessing Correctness and Genuine Novelty

Any claim that an LLM agent has solved a hard problem, such as the Navier-Stokes existence and smoothness problem, must pass two separate gates: the result must be correct, and it must be genuinely new. In research mathematics these questions are not interchangeable, and the difficulty of answering each on its own is precisely why agent-generated proofs attract scrutiny even when they seem to work. For a Millennium Prize problem the standard of evidence is a released, verifiable proof rather than an agent's confident assertion [16]. Reporting still therefore describes the Navier-Stokes outcome as a claim: OpenAI published a 166-page report

that constitutes the evidence, while the community treats it as provisional pending independent verification [16]. Online discussion frames the same event even more plainly, as 'OpenAI claims to have solved Navier Stokes Problem,' language that preserves the burden of proof [17].

The verification standard for hard mathematics

The standard that a verified proof satisfies is not met by another natural-language judgment from the same model family. The problem of verification using LLMs and proof assistants includes two subtasks, the first being autoformalizing the solution into Lean4 [18]. Autoformalization removes the ambiguity that lets errors slip past a human reader, forcing the informal argument to be encoded as machine-checkable lemmas so that any gap surfaces as a failure to formalize [18]. Correctness is then established by mechanically checking the formalized proof in a proof assistant rather than by trusting the generating model to have reasoned correctly [18].

Even a fully checked formal proof does not establish, however, that the result is novel. Benchmarks built around the same verification goal explicitly pair formal mathematics with contamination mitigation [19]. The underlying concern is that a system tested on problems already present in its training data cannot be said to be deriving the answer through genuine mathematical reasoning [19]. A candidate that merely reproduces a known proof from the corpus will pass a naive correctness check while still carrying the contamination signal that frontier-level evaluation is designed to flag [19].

Distinguishing genuine novelty from recombination

Novelty is harder to certify than correctness because it requires searching the entirety of existing mathematics rather than checking one argument's internal logic. The issues that distinguish research mathematics from contest solving include novelty, prior art, and ambiguity in problem statements [20]. A Millennium Prize problem has, by definition, no known solution, so a candidate that happened to coincide with already-published prior art could not legitimately claim to have resolved the problem [20]. Formal verification can prove that the conclusion follows from the premises, but it cannot by itself prove that the argument is new, because novelty is a claim about what already exists outside the proof itself [20]. Ambiguity in the problem statements themselves can further muddy the assessment, since different agents might interpret the same problem differently and produce results that are new relative to one interpretation yet irrelevant to the question actually posed [20].

Correctness and novelty as two distinct assessment gates

DIMENSION	CORRECTNESS GATE	GENUINE-NOVELTY GATE
Standard of evidence	A released, verifiable proof [16]	No existing prior art; the solution must not already be published [20]
Primary verification mechanism	Autoformalizing the solution into Lean4 and checking it in a proof assistant [18]	Searching existing mathematics for matching prior art [20]
Dominant failure mode	Informal steps that fool a reader but break during autoformalization [18]	Reproducing a known proof from training data instead of reasoning anew [19]

Note

Both gates must be cleared for an agent solution to count as a real result: a verifiable proof that also survives a prior-art check is what turns a bold claim into genuine novelty [16,20].

Other Agent Attempts at Hard Problems

Outside pure mathematics, LLM agents have been integrated with laboratory and research workflows, most notably in autonomous chemical research [21]. A Nature study on autonomous chemical research with large language models documented an agent that combines a language model with domain tools — reaction prediction, synthesis planning, and literature search — to run end-to-end scientific tasks, demonstrating that agents can now take on real research roles rather than only answering questions [21]. The same paper is candid about the limits, however, noting that addressing the complexities of software components and their interactions is crucial for integrating LLMs with laboratory automation [21]. That framing, in which the difficulty shifts from "can the model reason?" to "can the agent's tool stack and laboratory instruments be made to interoperate reliably?", is a recurring theme: the scientific value lives in the agent's closed loop with the world, not in the model's prompt. So even a chemically sound plan caps out at the reliability of the software and hardware it drives [21].

One cross-cutting observation is that the evidence base for these successes rests on benchmarks that may not capture what real science demands. A benchmarking study of AI agents for addressing scientific challenges argued that existing benchmarks rarely capture the complexity, heterogeneity, and extended reasoning required by actual scientific work [22]. This gap matters because a benchmark that rewards isolated problem-solving can make an agent look far stronger than it is when faced with a genuinely open-ended, multi-step problem. The very categories these agents attack — competition math, theorem proving, and formal-analysis benchmarks such as the MathMillenniumPrizeBenchmark — tend to be more structured than a live research program, so high scores do not automatically translate into solving a Millennium Prize problem wholesale [22]. In other words, the agents can look exceptional on the tests they are trained against, yet those tests omit much of what a real problem entails.

Across these cases, two patterns of success and failure recur. On the success side, the cases that perform best combine a specialized agent architecture — decomposition into sub-goals, tool use, and iterative refinement — with a conversational or collaborative structure that lets one agent critique another [23,24]. The 47-theorem output and the conversation-based competition math both rely on this kind of internal debate to generate and self-check candidates [23,24]. On the failure side, the repeated lesson is that generation is cheap and verification is expensive: claims that an agent has "solved" a hard problem — whether a Millennium Prize equation or a set of novel theorems — require external confirmation before they count, and the agents themselves do not close that loop on their own [24,25]. The chemistry case adds a third failure mode, showing that even a well-reasoned plan is only as useful as the automation it drives [21].

Alongside the celebrated Navier–Stokes episode, LLM-based agents have been pushed against a broad spread of genuinely difficult scientific and mathematical tasks, from laboratory chemistry to competition-level arithmetic and the formal proving of new theorems. The range of targets is notable precisely because each is judged against a hard external standard, so an agent's score is more than just fluency. Reading these cases together, three patterns recur, and the verification demand that defines the Navier–Stokes claim appears across every one of them [25].

Specialized agents in mathematics and chemistry

Outside pure mathematics, LLM agents have been integrated with laboratory and research workflows, most notably in autonomous chemical research [21]. A Nature study on autonomous chemical research with large language models documented an agent that combines a language model with domain tools — reaction prediction, synthesis planning, and literature search — to run end-to-end scientific tasks [21]. The study concluded that such agents can now take on real research roles rather than only answering questions [21]. The same paper is candid about the limits, however, noting that addressing the complexities of software components and their interactions is crucial for integrating LLMs with laboratory automation [21]. That framing, in which the difficulty shifts from "can the model reason?" to "can the agent's tool stack and laboratory instruments be made to interoperate reliably?", is a recurring theme across the whole literature. So even a chemically sound plan caps out at the reliability of the software and hardware it drives [21].

A benchmark gap and recurring failure modes

One cross-cutting observation is that the evidence base for these successes rests on benchmarks that may not capture what real science demands. A benchmarking study of AI agents for addressing scientific challenges argued that existing benchmarks rarely capture the complexity, heterogeneity, and extended reasoning required by actual scientific work [22]. Scientific work, by contrast, is heterogeneous and rarely fits a single prompt-and-answer template [22]. This gap matters because a benchmark that rewards isolated problem-solving can make an agent look far stronger than it is when faced with a genuinely open-ended, multi-step problem. The very categories these agents attack — competition math, theorem proving, and formal-analysis benchmarks such as the MathMillenniumPrizeBenchmark — tend to be more structured than a live research program, so

high scores do not automatically translate into solving a Millennium Prize problem wholesale [22]. In other words, agents can look exceptional on the tests they are trained against, yet those tests omit much of what a real problem entails.

Novelty vs. Recombination and Failure Modes

The claim that an LLM agent has found a novel solution to a hard problem, such as the Navier–Stokes existence and smoothness problem, raises a before-the-point question: is the output genuinely new, or is it a recombination of pre-existing patterns presented in a new arrangement? This distinction sits at the heart of the skepticism toward machine creativity. Creativity research almost universally evaluates a candidate output along two axes — novelty and value — which together form one of the most commonly cited criteria for judging whether a product is creative [26]. From the product perspective, what needs to be established is what makes a product creative, and that judgment is central to any fair evaluation of an LLM [26]. The skeptic's charge is that a combinatorial system can manufacture the appearance of novelty while never clearing the value bar, because it operates entirely over a distribution it was trained on and cannot genuinely exceed [27].

Recombination versus Insight

Combinatorial creativity reconfigures elements that already exist into new arrangements, and recent work frames it as a new frontier in generalization ability [28]. Under this reading, an LLM never introduces anything outside its training distribution; it merely resynthesizes what it has absorbed. The critic labels this outcome the illusion of machine creativity, because higher measured novelty does not by itself translate into real-world creative achievement [27]. Even when an agent is analyzed carefully, such analyses can uncover significant failure modes that betray the difference between synthesis and invention [27]. A combinatorial system can therefore produce outputs that look novel without possessing the understanding that would justify calling them innovative [27].

Combinatorial recombination versus genuine insight

ASPECT	COMBINATORIAL RECOMBINATION VIEW	GENUINE INSIGHT
Underlying process	Reconfigures elements drawn from the training distribution [27,28]	Is assumed to transcend the training distribution [27]
Standard evaluation criteria	Novelty paired with value [26]	Must be both new and genuinely valuable [26]
Empirical status for LLMs	Directly reproduced across sampled outputs [29]	Contested and only occasionally observed [27]

Documented Failure Modes

The combinatorial framing also provides tools for cataloging where these systems break down. Constraints can serve as minimal abstractions of failure modes, so one productive approach is to measure the proportion of error types among the creative artifacts that fail to satisfy their objectives [28]. A prime example is hallucination, in which the model confidently asserts something false: asking for the author of *Pride and Prejudice*, an LLM can erroneously produce 'Charles Dickens' rather than Jane Austen [30]. This failure is revealing precisely because it is fluent and confident, showing that plausibility is not a proxy for correctness and that novelty alone does not supply value [26].

A second failure mode is variability. Analyses of LLM creativity observe that outputs vary substantially, and that variability has important implications for both creativity research and practical applications [29]. Ignoring such variability risks misjudging the creative quality of a model, since a single striking example can be mistaken for a reliable capability [29]. That fragility undermines strong novelty claims: if a genuinely new and valuable result emerges only intermittently, the product-level evaluation that creativity research demands becomes difficult to satisfy consistently. The combined picture is that combinatorial recombination can be powerful, but it is exactly the kind of system that fails in predictable ways and that stops short of the innovation that novelty claims demand [27].

Note

The decisive test is not whether an output looks novel, but whether it clears the value bar: novelty and value are evaluated together, and a system that merely recombines a fixed distribution is expected to produce the former without the latter [26,27].

Implications for Creativity and Insight

The novelty-vs-recombination distinction raised in the previous section is not merely a technical concern; it opens a broader philosophical question about whether such agents exhibit genuine insight or merely mimicry, and what that implies about human creativity and mathematical discovery itself.

The creativity debate – novelty or mimicry?

The most direct engagement with this question comes from analyses of the current debate over whether machines can produce truly new ideas. Such analyses observe that whatever ideas an AI system generates exist within conceptual frameworks established by humans and draw from patterns in training data reflecting human creativity and prior knowledge [31]. In other words, the novelty is in the arrangement rather than in the conceptual vocabulary itself, which places the debate squarely inside the novelty-vs-recombination frame established earlier [31].

The concern is sharpened in research assessing whether generative artificial intelligence can make scientific discoveries similar to those made by humans [32]. The prevailing skeptical verdict is that generative AI lacks the human creativity required to achieve scientific breakthroughs comparable to the human benchmark, and that this gap is one of conceptual leap rather than of scale or compute [32].

What formal and competition successes reveal

The sharpest public illustration arrives from competition mathematics, where correctness is objectively measurable. DeepMind's AlphaProof and AlphaGeometry 2 became the first AI systems to reach a silver-medal standard at the 2024 International Mathematical Olympiad [33,34].

DeepMind itself described the achievement as a breakthrough in AI performance solving complex mathematical problems [33]. AlphaProof takes a formal approach to reasoning, and AlphaGeometry 2 is characterized as an even more competitive system [33]. The broader framing, captured in coverage of the future of mathematical reasoning, places these systems within the Lean proof-assistant ecosystem [34]. Yet this precise, machine-checkable setting is exactly why the result does not by itself settle the creativity question: a proof can be both correct and novel while remaining a recombination of training-distribution techniques, the very scenario the novelty-vs-recombination distinction warns about [31]. So a silver medal measures a speed-and-correctness benchmark rather than the generation of a genuinely new conceptual framework [33].

Assistant, not author

The practical implication that follows is a division of labor rather than a transfer of intellectual credit. Even in a domain as rigorous as mathematics, the assessment is that although AI can assist, it cannot generate the intellectual contributions to scientific research [35].

A scientific manuscript therefore retains human intellectual authorship even when AI assists in its drafting [35]. This positions the emerging future of mathematical reasoning as human-led collaboration rather than machine replacement [34]. Systems built around Lean and formal verification treat AI as a powerful tool for checking and exploring proofs while the human contributor remains the source of the ideas [34]. The trajectory, then, is one in which AI augments human creativity without yet substituting for it [32].

Insight versus novelty-from-patterns — where AI stands today

DIMENSION	HUMAN DISCOVERY	AI NOVELTY-FROM-PATTERNS
Origin of ideas	Establishes new conceptual frameworks [31]	Recombines training-data patterns [31]
Correctness	Reviewed by the human community	Mechanically verified in competition math [33]
Intellectual credit	Author of the insight [35]	Assists but does not generate the contribution [35]
Creativity bar	Paradigm of scientific breakthrough [32]	Silver-medal competition level [34]

If insight is understood as the capacity to introduce genuinely new conceptual frameworks, then the honest verdict is that current agents have not crossed that bar; they excel at recombination at silver-medal scale [31]. What they do signal is a shift in how human creativity will be exercised — less in grinding verification and more in framing, judging, and proposing [34]. The framing, judging, and proposing of genuinely new frameworks remains the distinctly human task.

Synthesis

The report's flagship case study, 'The Navier–Stokes Agent Attempt,' reads as a live instance of the tensions that the other sections diagnose. In that episode an LLM reportedly extended a human-created route to the Millennium Prize problem using heavy computation, though the claim stayed unconfirmed as of the original reporting [2,3,5]. 'Mechanisms of Novelty from Patterns' explains this recombination pattern, where an agent that seems to discover a new solution is recombining learned patterns past a scale threshold rather than inventing ex nihilo [6]. That reading sits in tension with 'Agent Architectures Enabling Novelty,' which argues that inference-time search and tool-augmented self-refinement let an agent reach configurations unreachable by pattern recall from training data alone [13,15]. The tension relaxes once the architecture is treated as a structural amplifier of emergence rather than an origin of brand-new ideas [15]. The raw novelty it supplies, however, still descends from recombination within the training distribution [6].

'Assessing Correctness and Genuine Novelty' resolves the agent's unconfirmed status by insisting that any claim must clear two separate gates: a released, verifiable proof and a clean prior-art check [16,20]. Because the Navier–Stokes report never received independent confirmation, it stayed confined to the first gate and so remained provisional [2,16]. A nearly identical vulnerability shapes 'Other Agent Attempts at Hard Problems,' where autonomous chemistry and competition-math successes share the property that generation is cheap but verification is expensive, and the agents do not close that loop themselves [21,24,25]. 'Novelty vs. Recombination and Failure Modes' generalizes the pattern into a conceptual distinction, warning that a combinatorial system can manufacture the appearance of novelty while never clearing the value bar [26,27].

The conclusion reaches its furthest point in 'Implications for Creativity and Insight,' which finds that current agents still lack the conceptual leap that would qualify as genuine insight, even as they clear silver-medal competition benchmarks [31,32,33].

Note

For an agent to clear a hard-problem claim, confidence or fluency is no substitute for a solution that is mechanically checked and proven free of existing prior art [18,20].

1. Build evaluations around a two-gate standard — machine-checked correctness plus a genuine prior-art check — instead of treating fluency or benchmark scores as a proxy for a result [18,20,26].
2. Read agentic architectures as amplifiers that expand search rather than as sources of conceptual novelty, and match evaluation to those recombination-level claims [6,13,15].
3. Reserve intellectual authorship for the humans who frame, judge, and propose, while using agents to assist with verification and exploration [34,35].

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